How Prediction Market Bots Actually Work

Most people think trading bots are magic. They're not. They're loops that read prices, compare them to a model, and place orders when the math is right. This guide breaks down the mechanics.

THE CORE IDEA

Prices are probabilities

On Polymarket, every market is a binary contract. A YES share priced at $0.42 means the market thinks there's a 42% chance the event happens. A NO share at $0.58 is the other side of that bet. If you believe the true probability is 55%, the YES share is underpriced by 13 cents. That's your edge.

> EXAMPLE: FINDING A MISPRICING
>
> Market: `"Fed cuts rates by July"` trading at `YES $0.42`.
> Your model estimates 55% probability based on inflation data + Fed dot plots.
> Expected value: `$0.55 - $0.42 = $0.13 edge per share`.
> At 500 shares, that's $65 in expected profit on a single position.

The catch: mispricings don't last. Other traders and bots are scanning the same markets. A 13-cent gap might close in minutes. The entire point of a bot is to detect and act on these faster than you can manually.

THREE STRATEGIES THAT WORK

Where the edge comes from

Market Making

Place buy and sell orders on both sides of a market. You profit from the spread between them. The key is managing inventory so you don't get stuck holding a losing position when the market moves.

Edge: spread capture, 1-4% per round trip

Arbitrage

Find the same event priced differently across correlated markets. If "Party X wins" is $0.60 on one market but the implied opposite is $0.35, that's a 5-cent risk-free gap. Buy the cheap side, short the expensive one.

Edge: price discrepancy, near risk-free

Directional

Build a model that estimates true probabilities better than the market. Use polling data, on-chain signals, or macro indicators. Buy when your model says the market is too low, sell when it's too high.

Edge: information advantage, variable

Most successful bots combine elements of all three. Market making generates steady small returns. Arbitrage captures risk-free opportunities when they appear. Directional bets are where the large gains come from, but they require a real information edge to sustain.

UNDER THE HOOD

Anatomy of a prediction market bot

Every trading bot, no matter how complex, follows the same five-stage pipeline. Data comes in, gets analyzed, triggers decisions, executes trades, and feeds results back into the system.

Data Feed
Stream prices, volume, order book from API

→

Scanner
Filter 800+ markets, score opportunities

→

Strategy
Decide position size, direction, entry price

→

Risk Check
Validate against limits before execution

→

Executor
Place orders, monitor fills, log results

The data feed is a WebSocket connection that streams live prices. The scanner runs your filters to surface only the markets worth looking at. The strategy layer decides what to do. The risk check is your safety net, making sure no single trade can blow up your portfolio. And the executor talks to the exchange.

THE DECISIONS THAT MATTER

What separates a toy from a real system

Position sizing

How much capital per trade? Kelly criterion is the gold standard: bet proportional to your edge divided by the odds. Most practitioners use half-Kelly to reduce variance.

Max drawdown rules

Set a hard limit on how much the portfolio can lose before the bot stops trading. 10-15% is typical. Without this, one bad streak wipes out months of gains.

Execution speed vs. cost

Limit orders save money but risk not getting filled. Market orders guarantee a fill but eat the spread. The right choice depends on how fast the opportunity is closing.

Correlation management

Betting on "Fed cuts rates" and "Recession by Q4" might feel like two trades. They're often the same bet in disguise. Your bot needs to track cross-market exposure.

The most common mistake

Building a bot that works in backtesting but fails live. The gap is almost always **execution assumptions**. In backtests, orders fill instantly at the exact price you wanted. In production, there's slippage, latency, and other bots competing for the same opportunity. Always backtest with realistic fill models, then paper-trade before going live with real capital.

Want to build this yourself?

Book a free 15-min call. We'll assess your background and map out where to start.

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